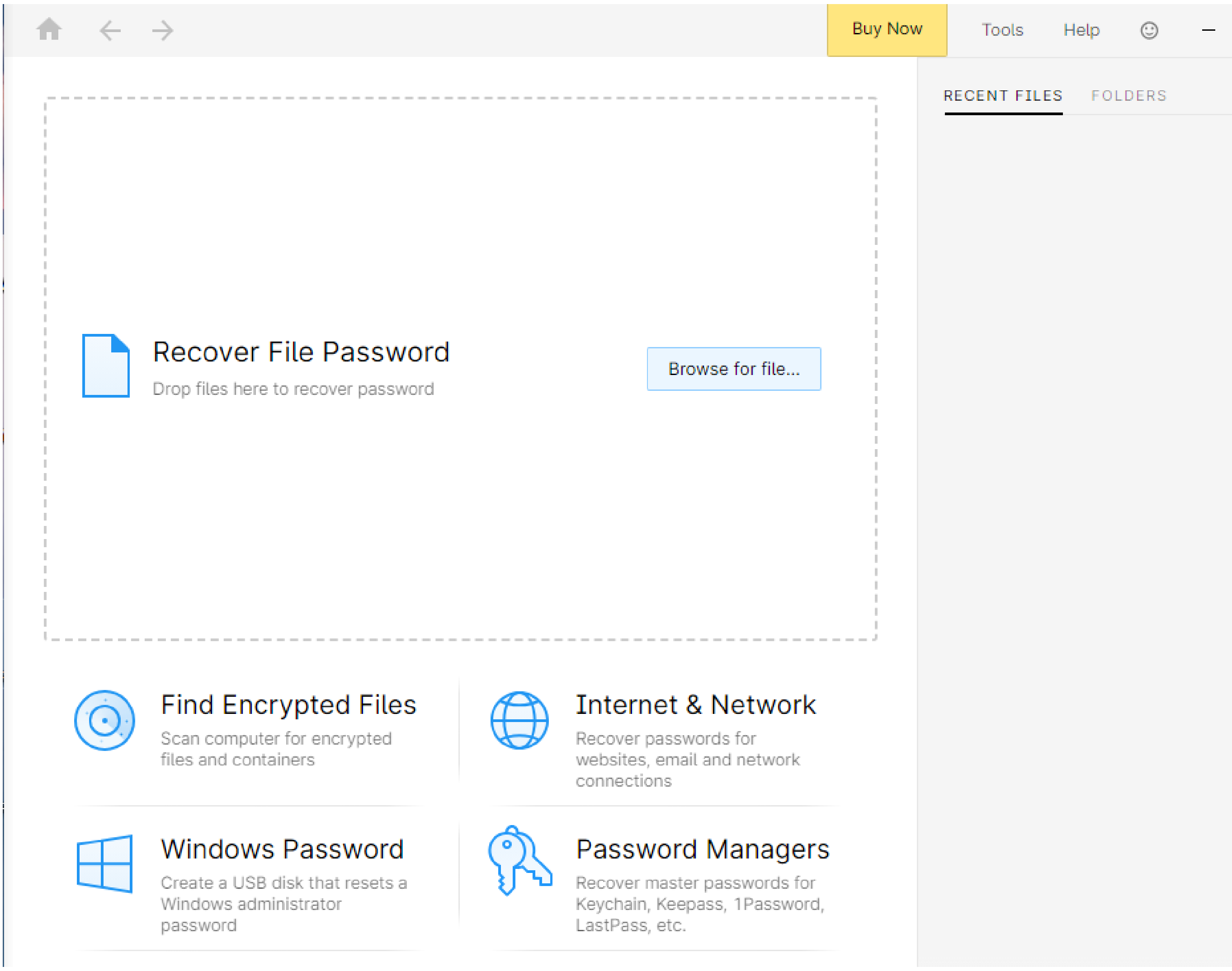
Cyber Forensics - 24CY611
Lab 4 - Defeating Anti-forensics Techniques

1. Cracking Application Password

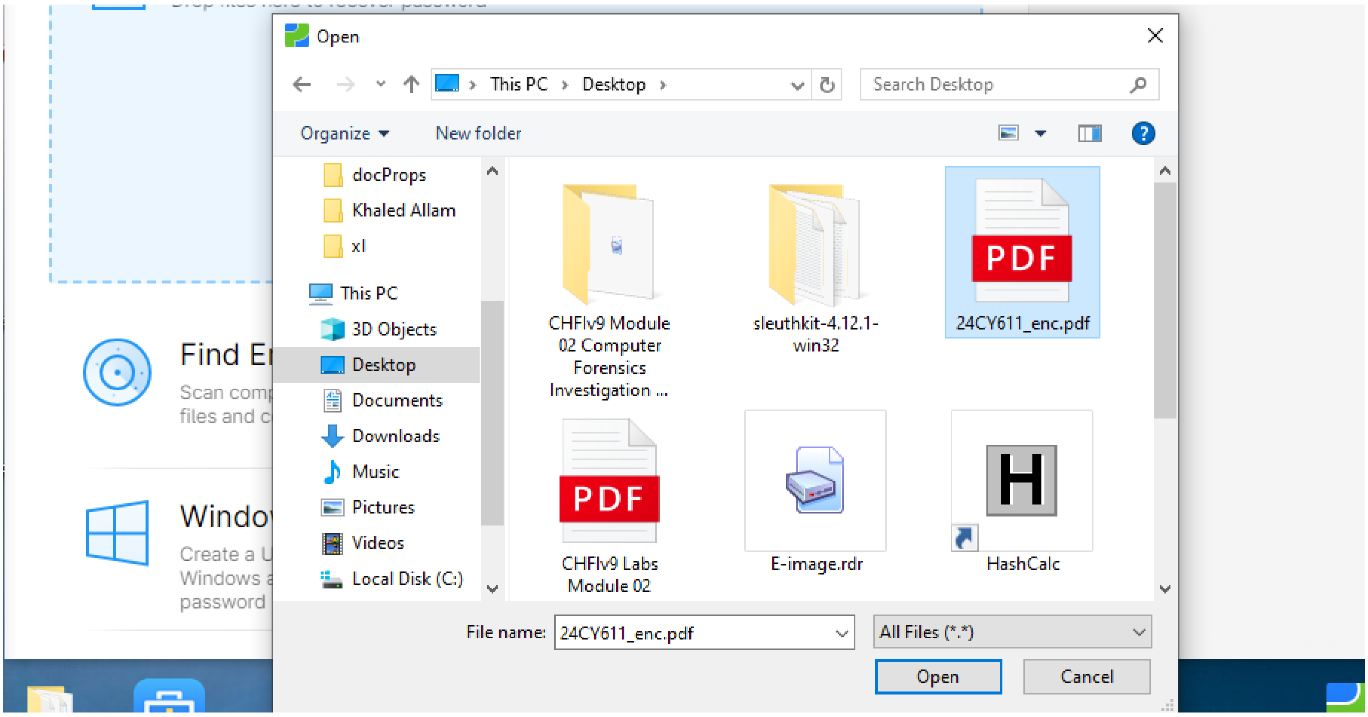
Passware kit

Passware Kit is a commercial password recovery tool that unlocks files, documents, and encrypted devices. It supports various formats and uses techniques like dictionary, brute-force, etc..

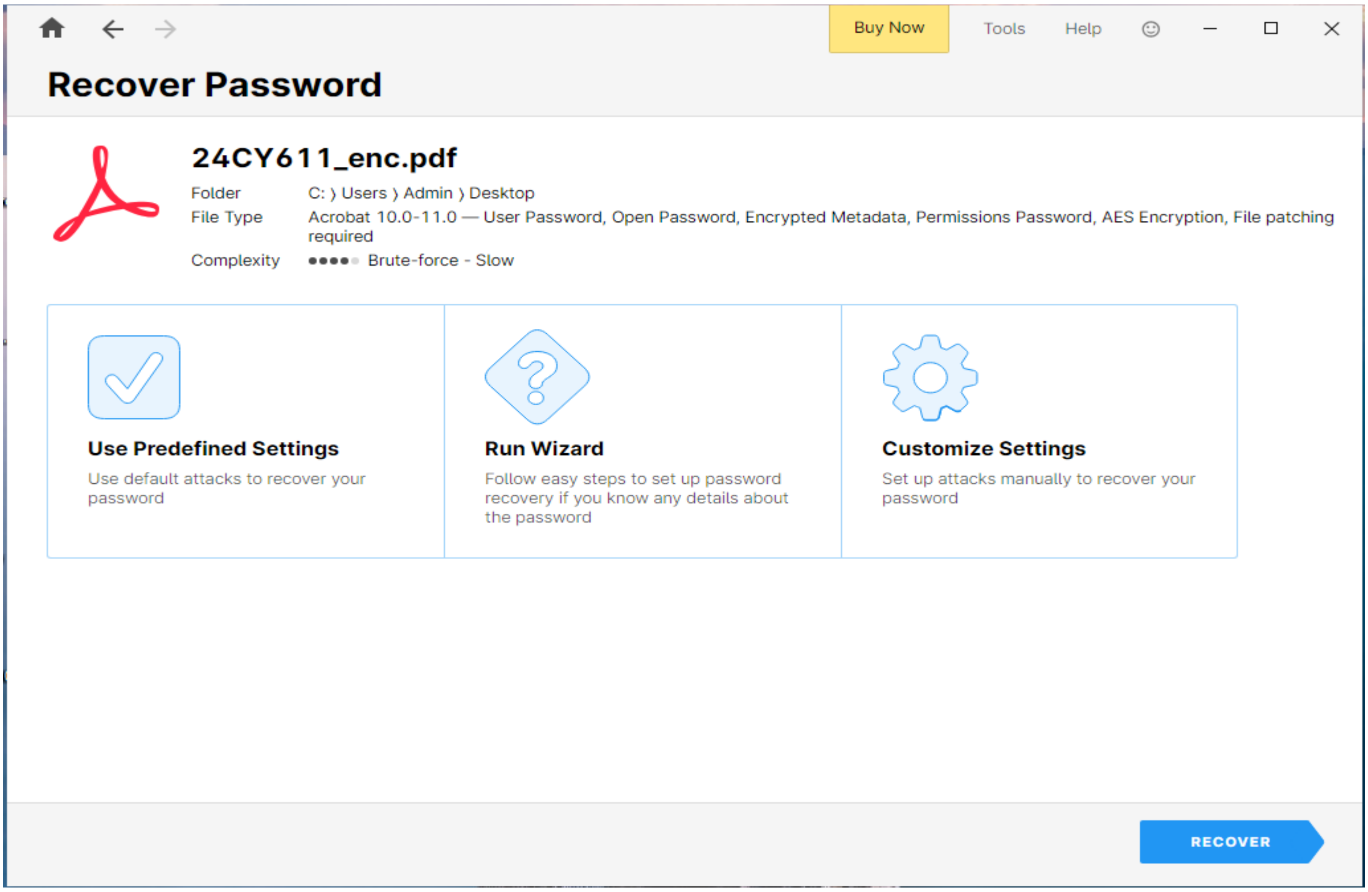
Open passware kit and select the file through browse for file.



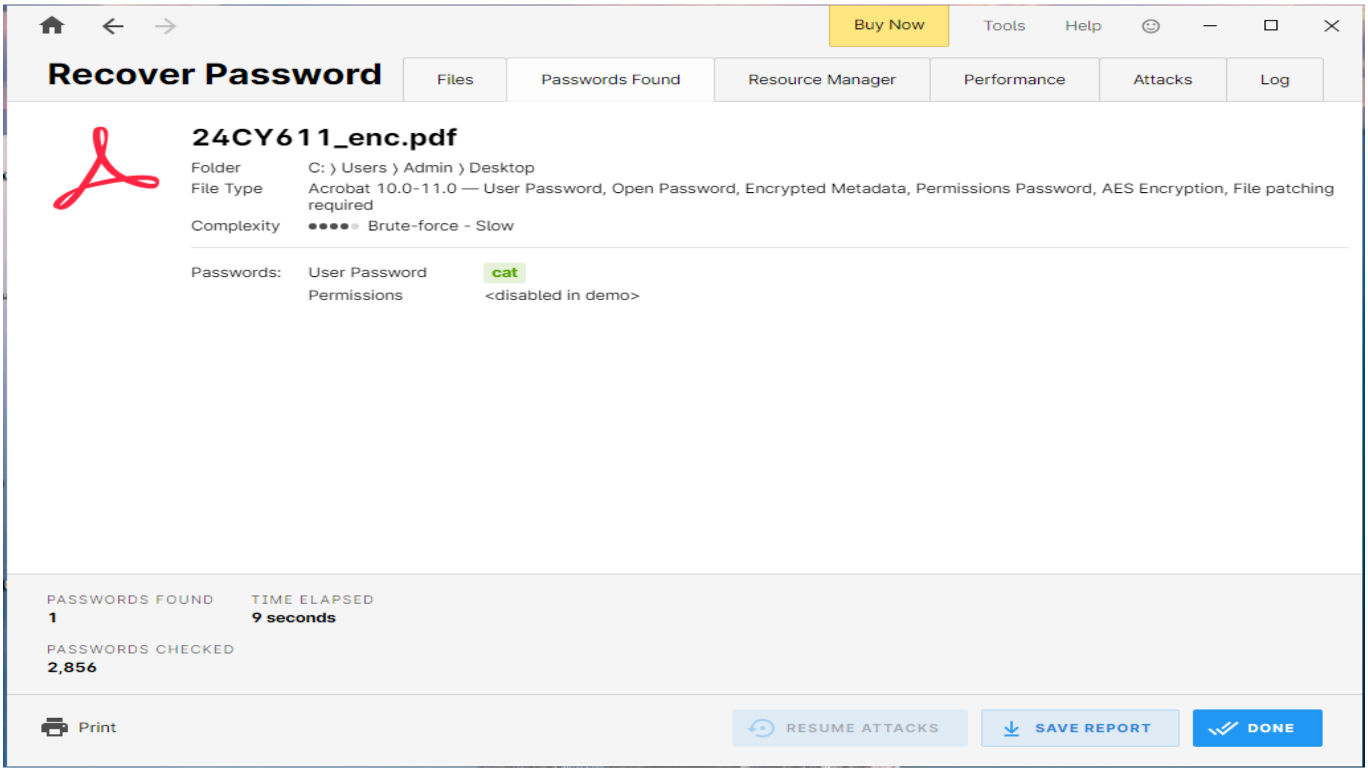
Select the file and click open



Then click the Recover button at the bottom right to crack the password.



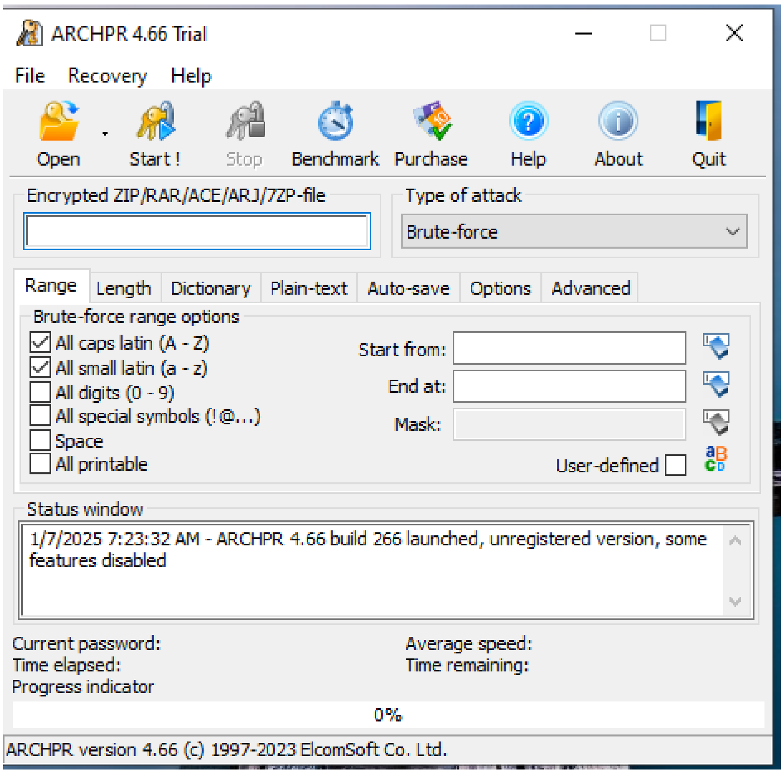
Recover password window will display the cracked password.



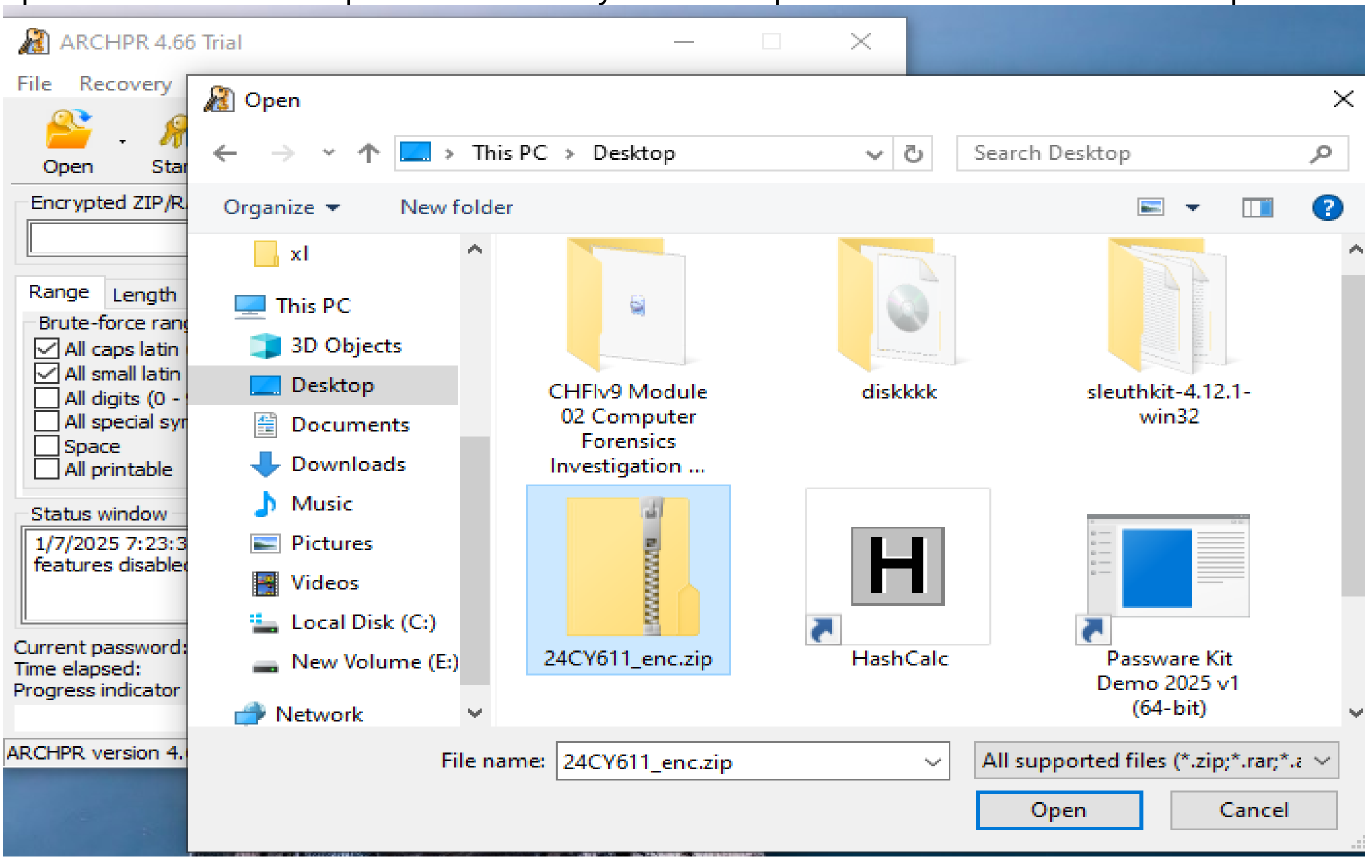
Recoverd password : cat

Advanced Archive Password Recovery

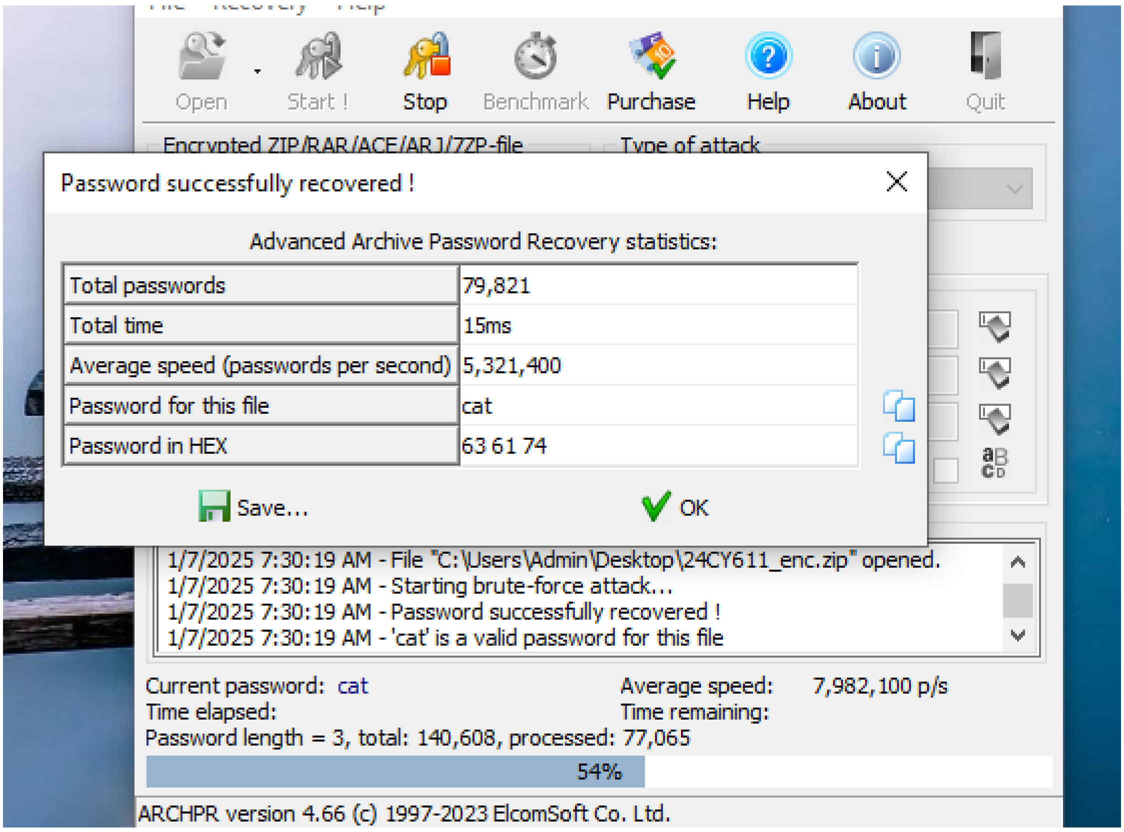
Advanced Archive Password Recovery is a tool for recovering lost passwords from ZIP, RAR, and other archive formats using brute-force, dictionary, and known plaintext attacks efficiently.



open Advanced archive password recovery and click open to load the file. Then click Open



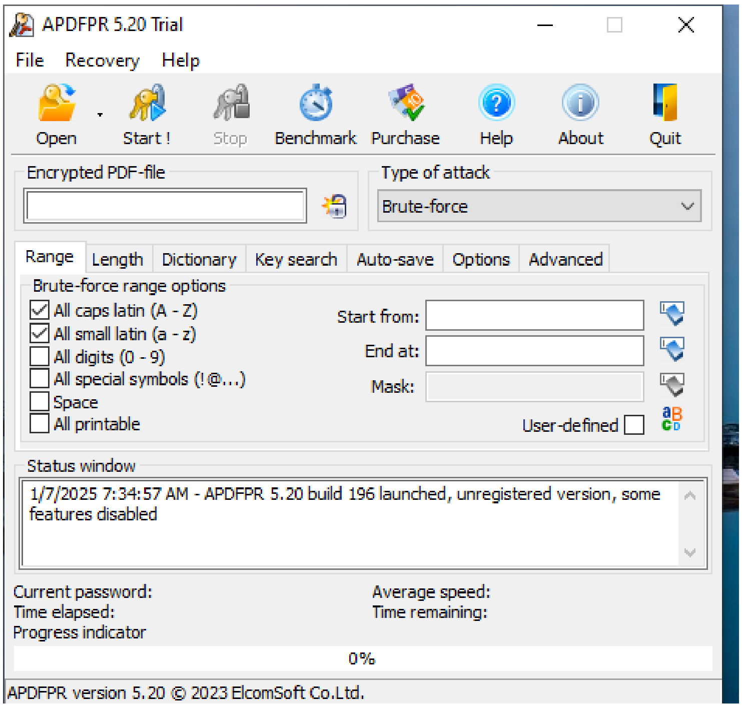
This shows that the password is successfully recoverd . Recovered password is cat.

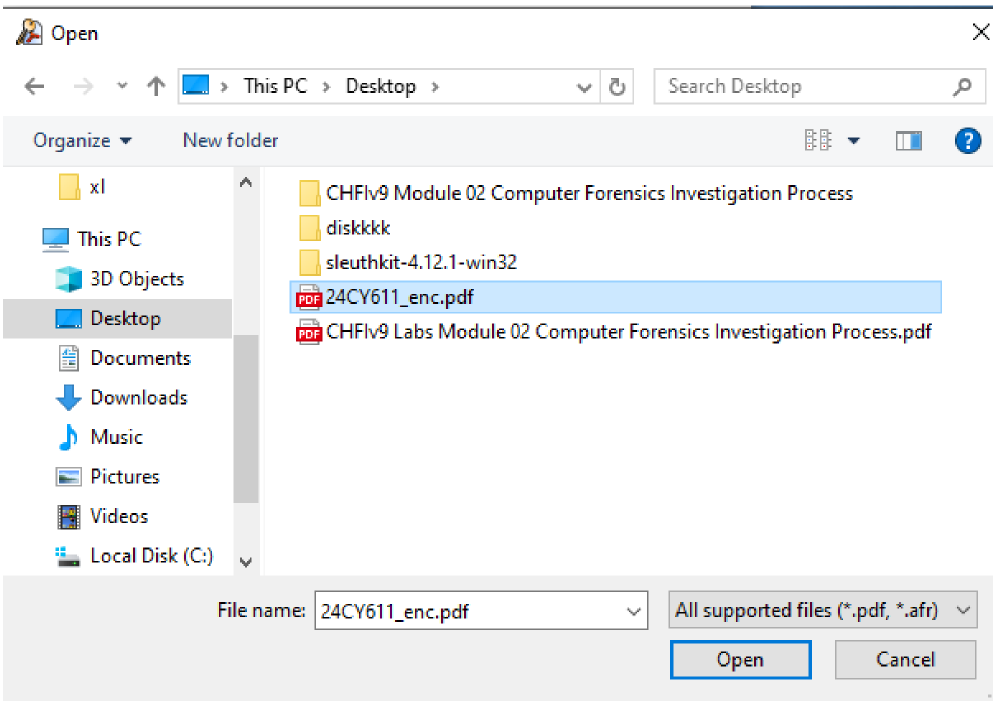


Advanced PDF Password Recovery

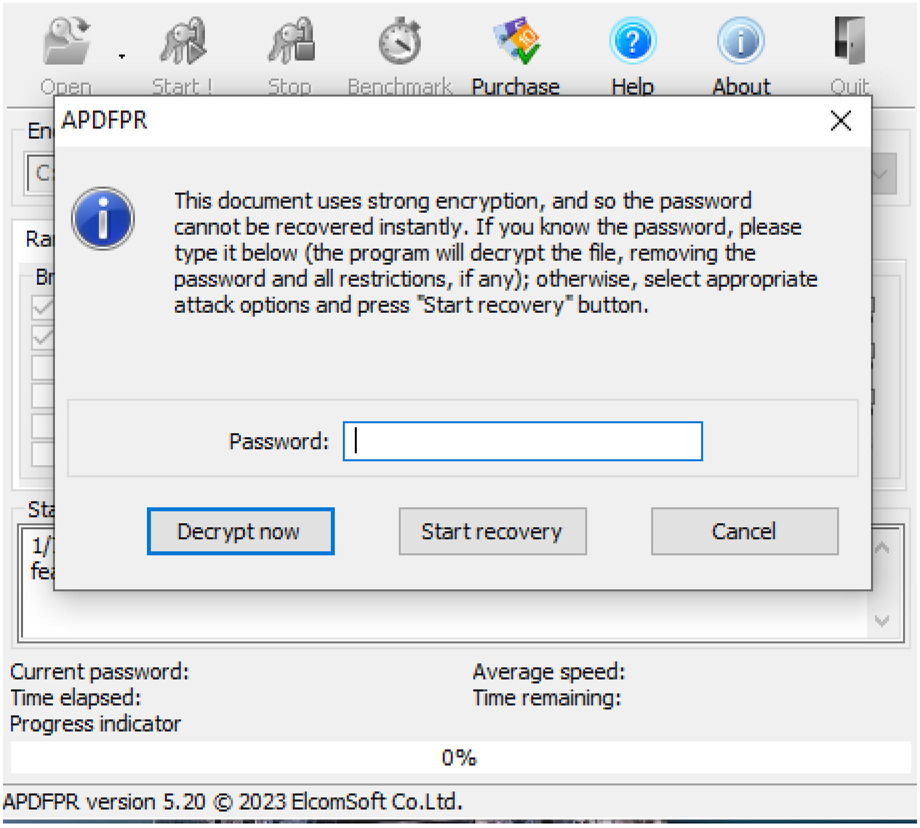
Advanced PDF Password Recovery is a tool used to unlock password-protected PDF files by removing restrictions or recovering user passwords through brute-force, dictionary, and key search attacks.

Open Advanced PDF Password Recovery and click Open button to load the file. Then click Open.

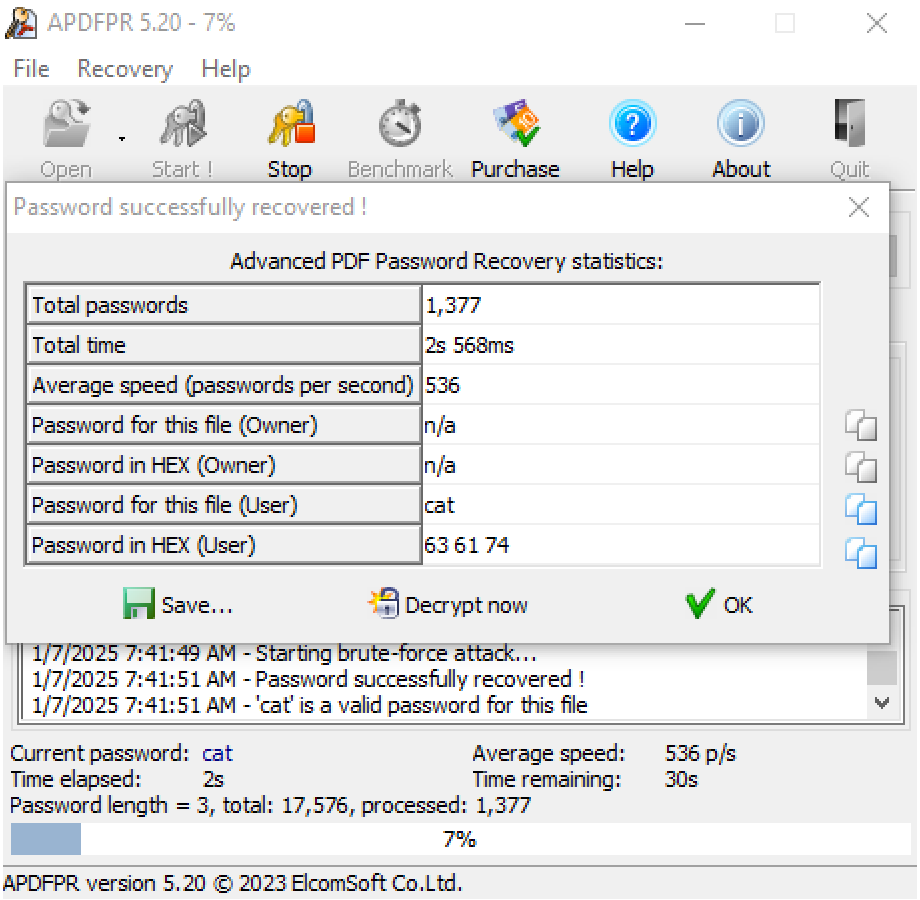




Click start recovery



This displays that the password is successfully recoverd . Recovered password is cat.



John the Ripper password cracker

John the Ripper is a password-cracking tool used for security testing. It performs brute-force, dictionary, and custom attacks to identify weak passwords across various hash types.

Open terminal and use the command pdf2john <pdffile> to get the hash of the password

Then use the john command to bruteforce the hash with wordlist to find the actual value of the hash.

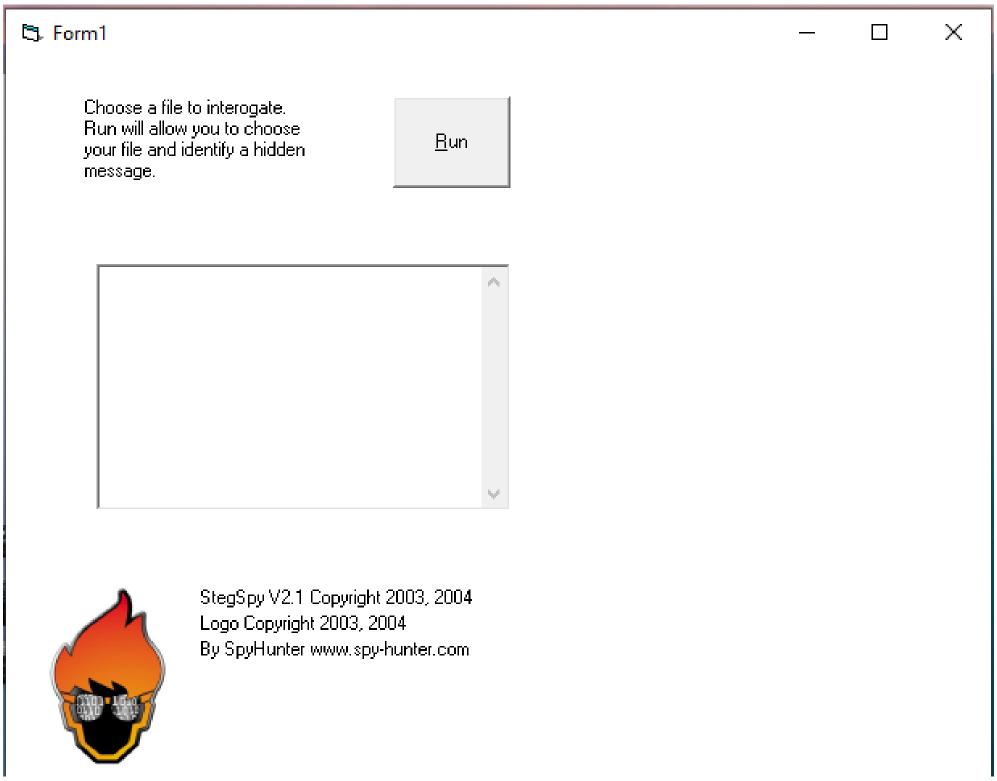
```
┌──(kali㉿kali)-[~/Desktop]
└─$ pdf2john 24CY611_enc.pdf
24CY611_enc.pdf:$pdf$5*6*256*-4*1*16*a8868818be6e7745999a2849528b8fa3*48*eb16d3d0bed5ff8ec2bb0e2da
aabd09b3ea7081d078fc0e74688840c63c613b7dfc351fd47c0fa*48*369fc01fe433db4bbaf6d97d48be832bd104159d9
49de3cf22640c9b81e0e0f1869217a6151ca4*32*4d58eacaf654f07a3885504941c8b96956a18c6efc0ee19b51dab05c0
bf0b257aecc0cc93ddf285701171c04aa3ee0b5ff30a14f1238d03ff3

┌──(kali㉿kali)-[~/Desktop]
└─$ pdf2john 24CY611_enc.pdf > passhash.txt

┌──(kali㉿kali)-[~/Desktop]
└─$ john passhash.txt
Using default input encoding: UTF-8
Loaded 1 password hash (PDF [MD5 SHA2 RC4/AES 32/64])
Cost 1 (revision) is 6 for all loaded hashes
Will run 2 OpenMP threads
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
Almost done: Processing the remaining buffered candidate passwords, if any.
Proceeding with wordlist:/usr/share/john/password.lst
cat              (24CY611_enc.pdf)
1g 0:00:00:07 DONE 2/3 (2025-01-07 23:03) 0.1277g/s 1764p/s 1764c/s 1764C/s 123456..gandalf
Use the "--show --format=PDF" options to display all of the cracked passwords reliably
Session completed.
```

2. Detecting Steganography

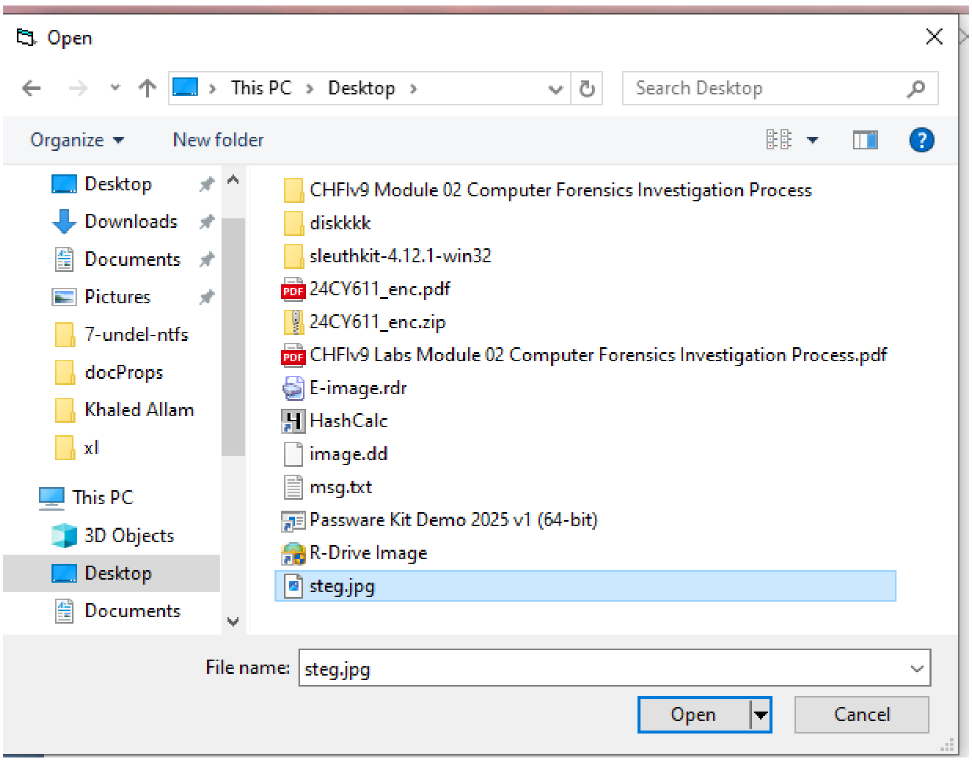
Stegspy

StegSpy is a steganalysis tool designed to detect hidden messages in image files by analyzing them for signs of steganography. It helps identify potential data concealment in digital images.

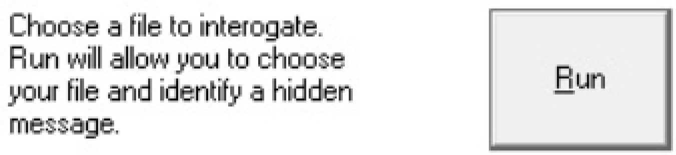
Right click on StegSpy2.1.exe and run as administrator



Click run button and select the steg image



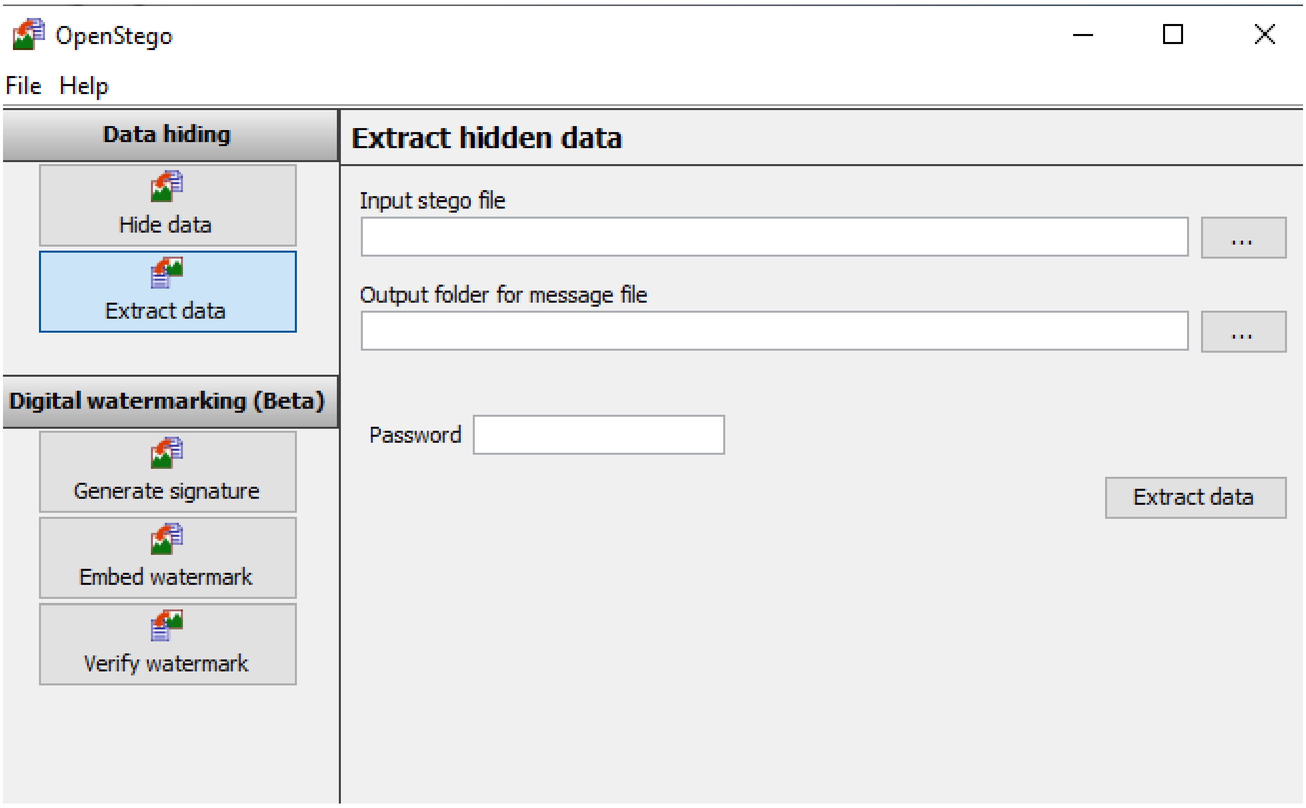
This displays the image has hidden message

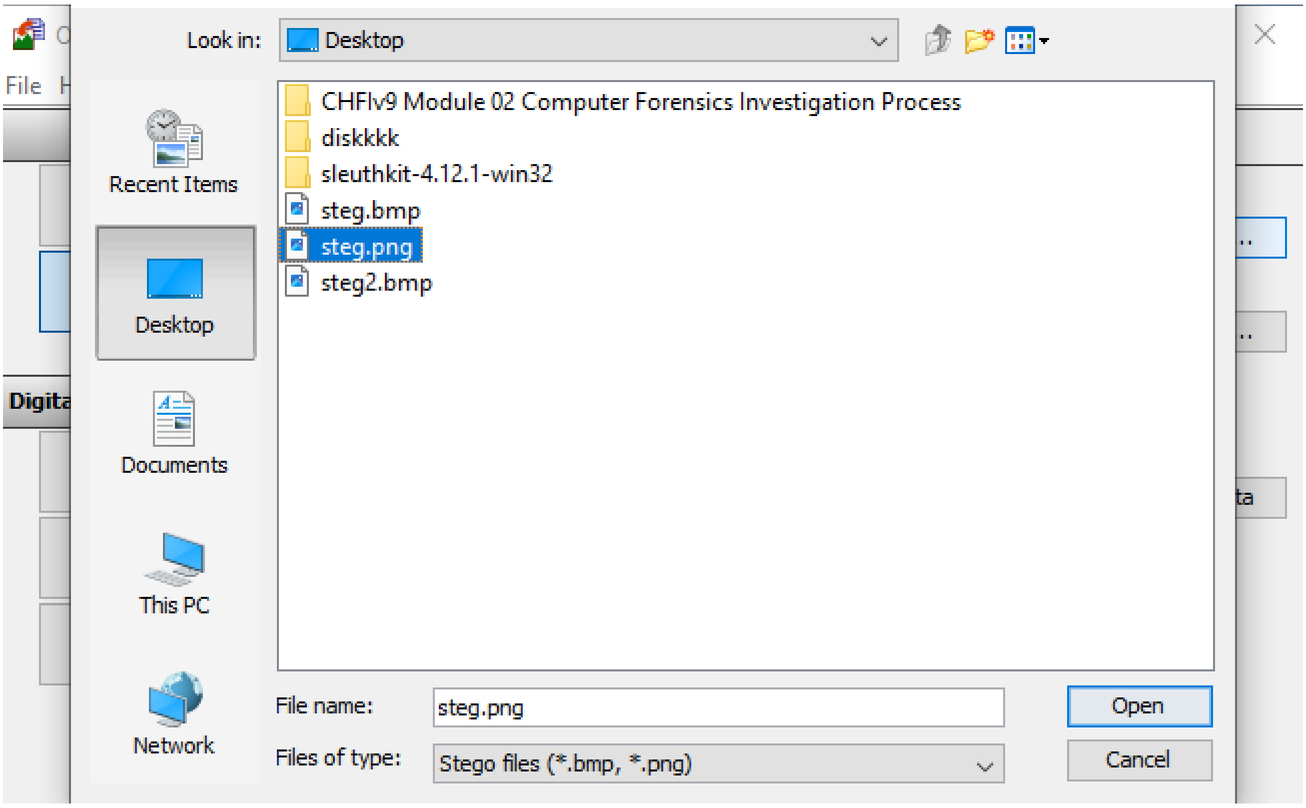


Openstego

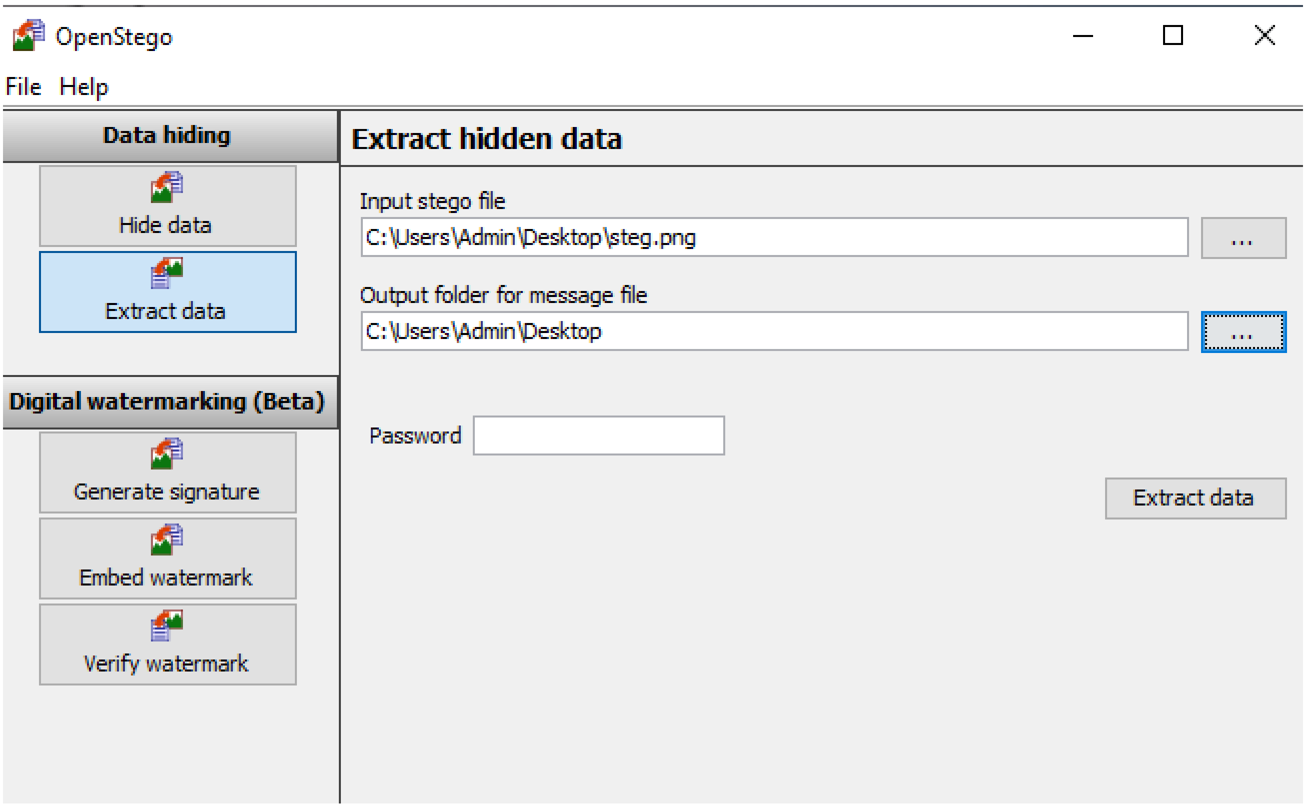
OpenSteg is a steganography tool that allows users to hide confidential data within image or audio files. It focuses on secure data embedding and extraction to protect sensitive information.

Open Openstego select the input stego file and click open





Then select the output folder and click extract



This displays that the hidden data has been successfully extracted from the steg image and savedd in a msg.txt file

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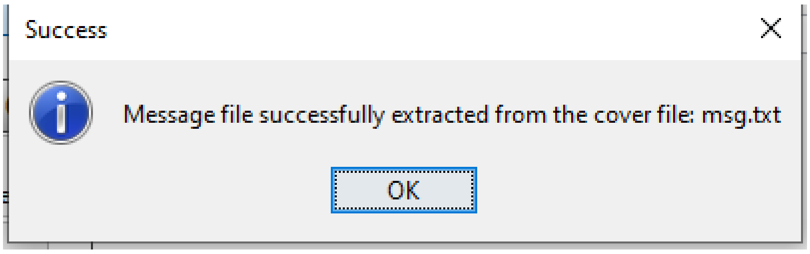
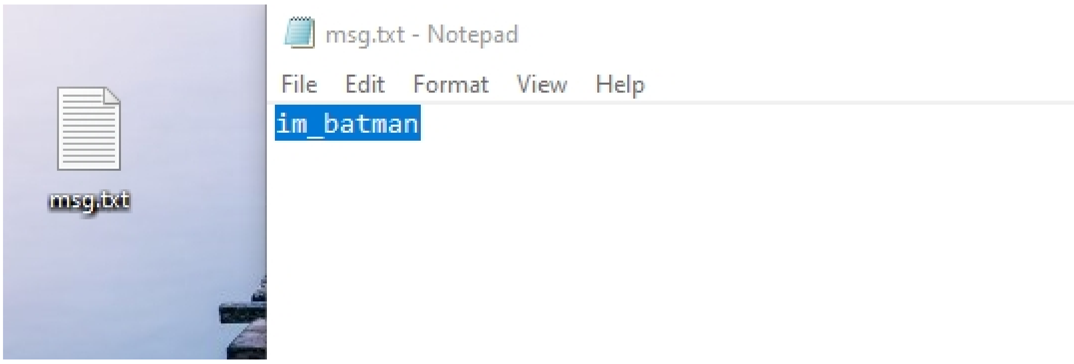
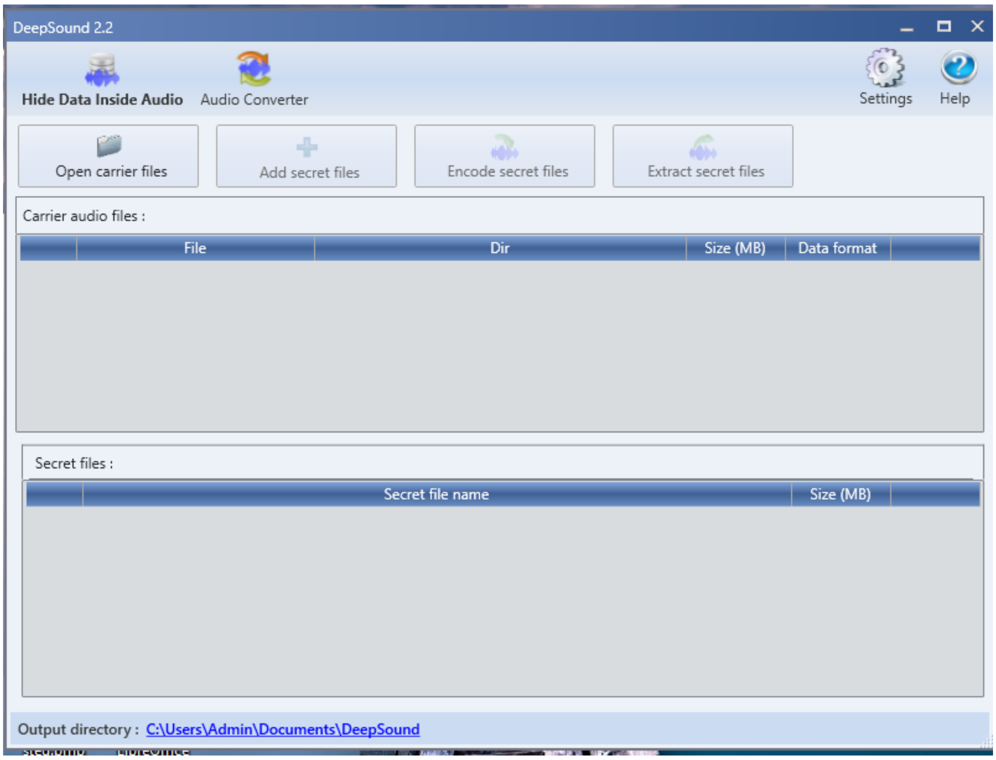




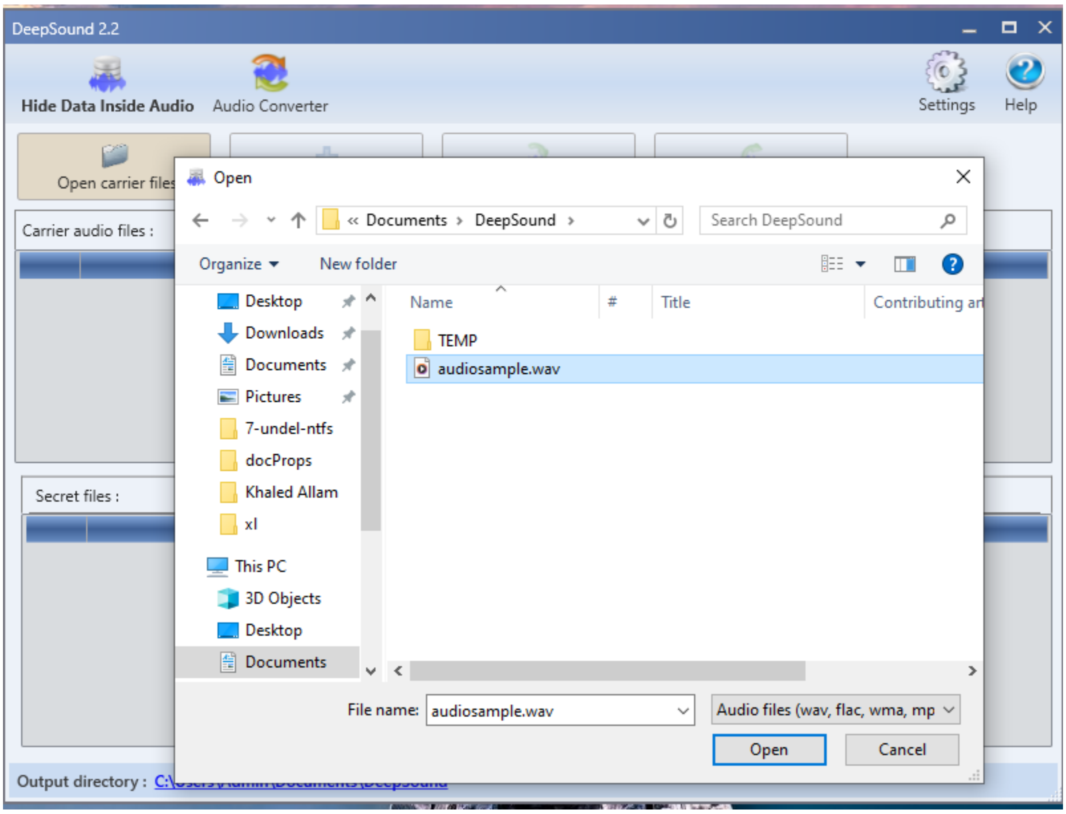
DeepSound

DeepSound is an audio steganography tool that hides secret data inside audio files (WAV, FLAC) while preserving sound quality. It can also encrypt hidden data for added security.

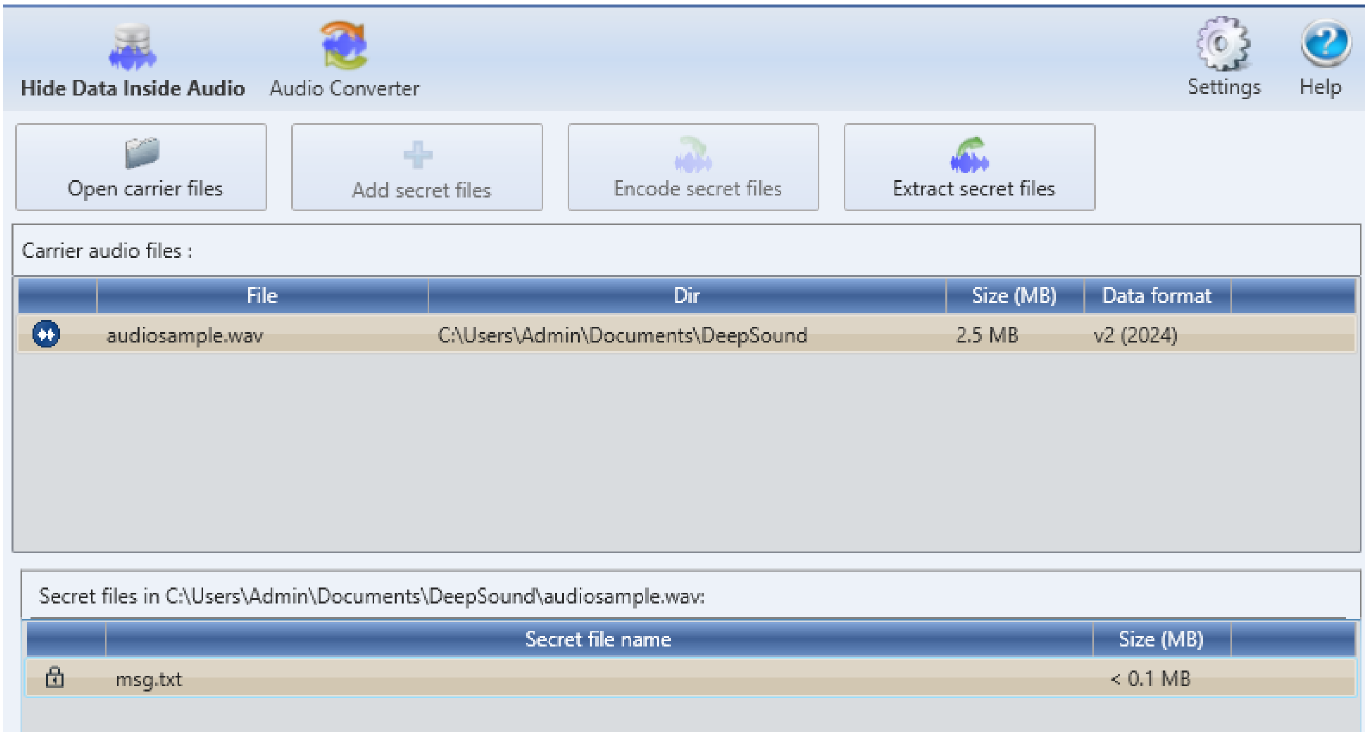
Open DeepSound



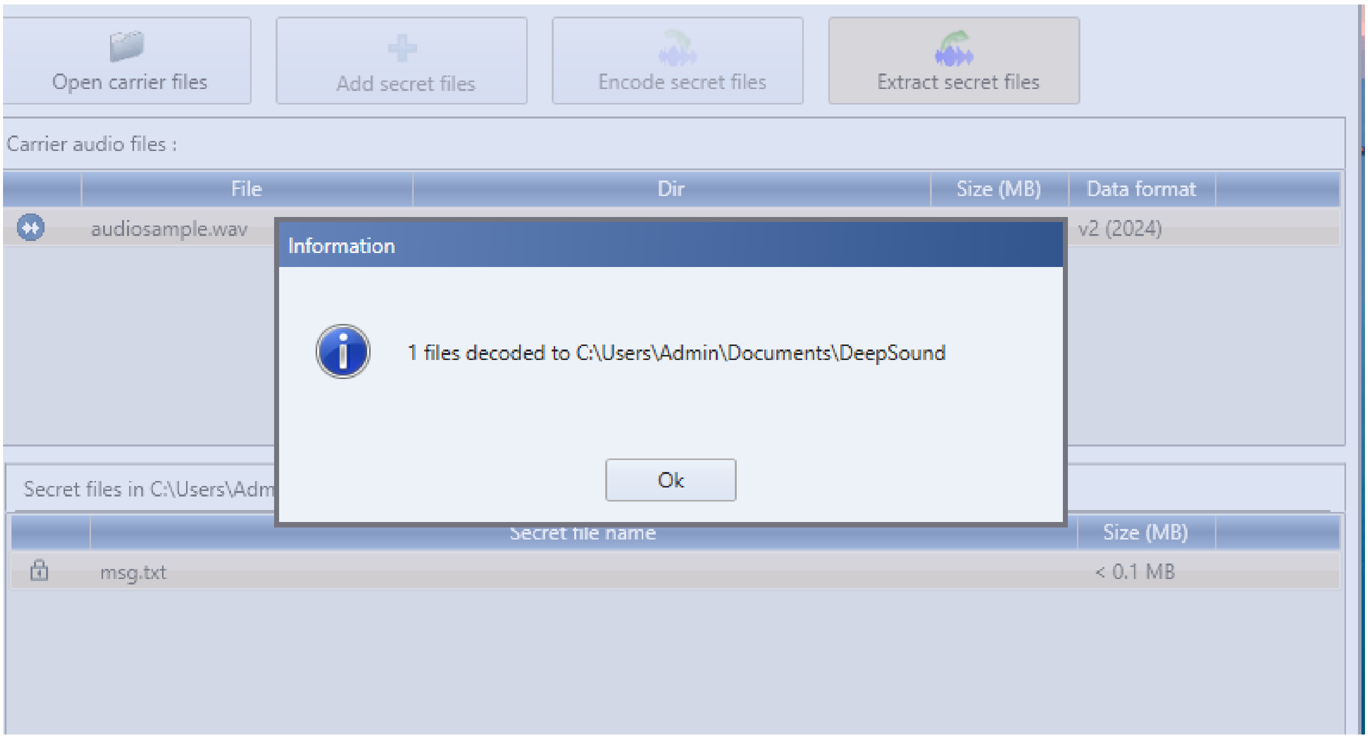
Click open career files and select the audio file and click open.



This displays that there is a secret file encoded to the audio file.



Click on the extract secret files



This is the secret message encoded in the audio file.